

Application of a hybrid of genetic algorithm and particle swarm optimization algorithm for order clustering

R.J. Kuo^{a,*}, L.M. Lin^b

^aDepartment of Industrial Management, National Taiwan University of Science and Technology, No. 43, Section 4, Kee-Lung Road, Taipei 106, Taiwan, ROC

^bAU Optonics Corporation, No. 1, Li-Hsin Road. 2, Hsinchu Science Park, Hsinchu 300, Taiwan, ROC

ARTICLE INFO

Article history:

Received 5 January 2010

Accepted 9 May 2010

Available online 17 May 2010

Keywords:

Clustering analysis

ART2 neural network

Particle swarm optimization algorithm

Genetic algorithm

ABSTRACT

This study proposes an evolutionary-based clustering algorithm based on a hybrid of genetic algorithm (GA) and particle swarm optimization algorithm (PSOA) for order clustering in order to reduce surface mount technology (SMT) setup time. Simulation results via Iris, Glass, Vowel and Wine benchmark data sets indicate that the proposed evolutionary-based clustering algorithm is more accurate than the GA-based and PSOA-based clustering algorithms. In addition, the model evaluation results which use order information provided by an international industrial personal computer (PC) manufacturer show that the proposed algorithm is also superior to GA-based and PSOA-based clustering algorithms. Through order clustering, scheduling orders that belong to the same cluster together can reduce production time as well as machine idle time.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

In the industrial personal computer (PC) industry, printed circuit board (PCB) assembly is a fundamental manufacturing process, in which surface mount technology (SMT) plays a very important role. By applying an SMT production system, not only are there more components on the limited space of a PCB, but production efficiency and product stability are also enhanced. However, the high mix low volume production style presents difficulties for SMT production systems. In this kind of production system, production line-changes create very serious bottlenecks since before, production labor must prepare for the materials and also bind the materials prior to line-changes, which is very time consuming. If production labor cannot complete these tasks before the next order starts, it causes both idle time for these expensive machines and decreases production capability utilization. Currently, this problem is frequently encountered in SMT production system, and the only way it can be handled is to prepare for more materials and increase the number of binding labors. An alternative is to cluster the orders before scheduling and then schedule similar orders together in order to reduce the material-binding time.

Clustering analysis is the process of identifying natural groupings or clusters within multidimensional data, based on some similar measures, like Euclidean distance [27]. Basically, as one of the most primitive activities of human beings, clustering plays an important and indispensable role in the long history of human development. In order to learn about a new object or understand a new phenomenon, people

always try to seek features that can describe it, and further compare it with other known objects or phenomena, based on similarity or dissimilarity, generalized as proximity, according to certain standards or rules [50].

Thus, this study proposes an evolutionary-based clustering algorithm based on a hybrid of genetic algorithm (GA) and particle swarm optimization algorithm (PSOA) for order clustering. It is named hybrid of GA and PSOA (HGAPSOA). Order features are characterized using bills of materials (BOMs) for all products. Through order clustering, the orders with similar materials are grouped together. The production engineer can schedule the similar orders together based on clustering results in order to reduce the setup time for SMT line-change operations in the industrial PC industry.

For assessing the proposed HGAPSOA, four benchmark data sets are employed, including iris, wine, vowel, and glass. The simulation results show that the proposed HGAPSOA outperforms the GA-based and PSO-based algorithms proposed by previous studies. In addition, evaluation results using data provided by an international industrial PC manufacturer also show that the proposed HGAPSOA is more accurate than other algorithms. By integrating the proposed algorithm with a scheduling method, this study's method can dramatically reduce SMT production system setup time and increase the utilization of machine capability.

The remainder of this study is organized as follows. Section 2 briefly presents the necessary background, including discussion of clustering analysis and particle swarm optimization algorithms, while the proposed order clustering system is explained in Section 3. Sections 4 and 5 discuss the simulation results using four benchmark data sets and the evaluation results for a real-world problem, respectively. Finally, concluding remarks are made in Section 6.

* Corresponding author. Tel.: +886 2 27376328; fax: +886 2 27376344.

E-mail address: rjkuo@mail.ntust.edu.tw (R.J. Kuo).

2. Background

This section briefly presents the general backgrounds of clustering analysis and PSO.

2.1. Clustering analysis

Clustering analysis partitions data into a certain number of clusters. Most researchers describe a cluster by considering internal homogeneity and external separation, i.e., patterns in the same cluster should be similar to each other, while patterns in different clusters should not [50].

Most clustering algorithms are based on two popular techniques known as hierarchical and partitional clustering [24]. Hierarchical clustering algorithms organize data into a hierarchical structure according to the proximity matrix. The results of hierarchical clustering are usually depicted by a binary tree or dendrogram. Hierarchical clustering algorithms are mainly classified as agglomerative methods and divisive methods. Many algorithms have been used in this clustering method, for example, CURE [20], BIRCH [51], ROCK [21], CHAMELEON [28], DIANA [29], AUTOCLUST [16] and AMOEBA [15].

In contrast to hierarchical clustering which yields a successive level of clusters by iterative fusions or divisions, partitional clustering assigns a set of objects into clusters with no hierarchical structure. These algorithms try to minimize certain criteria, like square error function, and can therefore be treated as optimization problems. Partitional clustering aims to optimize cluster centers, as well as the number of clusters [23]. Partitional clustering can be further classified into supervised and unsupervised clustering algorithms. The main difference between these clustering algorithms is that supervised clustering algorithms need to specify the number of clusters. They are discussed in more detail as follows.

2.1.1. Supervised clustering algorithms

The most widely used supervised clustering algorithm is the K-means algorithm [17]. The K-means algorithm starts with k cluster centroids, which are initialized to random values. Furthermore, each point of the data set is assigned to the cluster closest to it. The centroids are then recalculated according to their associated points. This process is repeated until convergence is reached. Latter and Bezdek [5] developed a fuzzy version of the K-means algorithm, called the fuzzy C-means algorithm. The object can belong to all of the clusters with a certain degree of membership. Other modifications of fuzzy C-means algorithms are the possibilistic-means clustering algorithm [32], fuzzy c-shells [6], and hierarchical unsupervised fuzzy clustering [19].

In addition, evolutionary algorithms are also employed to improve the performance of K-means. Clustering can be regarded as a category of optimization problems that uses evolutionary algorithms, like genetic algorithms (GA), particle swarm optimization algorithms (PSOAs) or ant colony optimization algorithms. Many techniques support this method, such as the genetically guided algorithm [22], the genetic K-means algorithm [31], Tabu search clustering [1], the ant colony clustering algorithm [35], and the particle swarm K-means optimization algorithm [37]. But the main drawback of these clustering algorithms is the process of parameter selection.

2.1.2. Unsupervised clustering algorithms

Regarding unsupervised clustering algorithms, artificial neural networks are the most representative. An artificial neural network (ANN) is a system that has been derived through models of neurophysiology. In general, they consist of a collection of simple nonlinear computing elements whose inputs and outputs are linked to form networks. ANN-based clustering has been dominated by self-organizing feature maps (SOM) and adaptive resonance theory (ART). Some important representative algorithms include SOM [30], ART1 [7], ART2 [8], and fuzzy ART [9].

SYNERACT [26], an alternative approach to ISODATA [3], combines K-means with hierarchical descending approaches. It is not necessary to specify the number of clusters. Based on K-means, other unsupervised clustering algorithms have also been developed, like X-means [46] and G-means [23].

Gath and Geva [18] proposed an unsupervised clustering algorithm based on the combination of fuzzy C-means and fuzzy maximum likelihood estimation. Lorette et al. [41] proposed an algorithm based on fuzzy clustering to dynamically determine the number of clusters in a data set.

Lee and Antonsson [38] applied an evolution strategy to dynamically cluster a data set. The evolution strategy they proposed implemented variable length genomes to search for both the centroids and K. In addition, Omran et al. [45] proposed a new dynamic clustering approach (DCPSO) based on particle swarm optimization algorithm.

2.2. Particle swarm optimization algorithm (PSOA)

PSOA shares many similarities with evolutionary computation techniques such as GAs. Systems are initialized with a population of random solutions and searches for optima by updating generations. However, unlike GA, PSOA has no evolutionary operators, such as crossover and mutation. In the PSOA, the potential solutions, called particles, move through the problem space by following the current optimum particles.

PSOA was first designed by Kennedy and Eberhart in 1995. In the original PSOA, particles are represented as $X_i = (x_{i1}, x_{i2}, \dots, x_{iD})$, which represents a potential solution to a problem in D -dimensional space. Each particle keeps a memory of its previous best position P_{best} , and a velocity along each dimension, represented as $V_i = (v_{i1}, v_{i2}, \dots, v_{iD})$. At each iteration, the position of the particle with the best fitness value in the search space, designated as g , and the P vector of the current particle are combined to adjust the velocity along each dimension, and that velocity is then used to compute a new position for the particle [14]. This method can be divided into the GBEST and LBEST versions [14], whose main difference is their definition of what is the best. In the GBEST version, the particle swarm optimizer keeps track of the overall best value, and its location, obtained thus far by any particle in the population, which is called g_{best} (P_{gd}). For the LBEST version, in addition to g_{best} , each particle keeps track of the best solution, called l_{best} (P_{gd}), attained within a local topological neighborhood of particles. However, the particle velocities in each dimension are held to a maximum velocity, V_{max} , and the velocity in that dimension is limited to V_{max} . The updating rule is as follows:

$$V_{id}^{new} = V_{id}^{old} + c_1 \cdot rand_1 \cdot (P_{id} - X_{id}) + c_2 \cdot rand_2 \cdot (P_{gd} - X_{id}) \quad (1)$$

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new} \quad (2)$$

where c_1 and c_2 determine the relative influence of the social and cognitive components (learning factors), while $rand_1$ and $rand_2$ denote two random numbers uniformly distributed in the interval [0,1].

In the original version, the maximum velocity V_{max} serves as a constraint to control the global exploration ability of a particle swarm. As stated above, a larger V_{max} facilitates global exploration, while a smaller V_{max} encourages local exploitation. The concept of an inertia weight was developed to better balance exploration and exploitation in order to eliminate the needs of V_{max} . The inclusion of an *inertia weight* in the particle swarm optimization algorithm was first reported by Shi and Eberhart in 1998 [47]. The authors proposed two definitions of inertia weight: the first one defines inertia weight as a positive constant; the second defines it as variably decreasing with time. This is because the larger inertia weight has more exploration ability at the beginning to find a good seed, and the later small inertia weight has more exploitation ability to search the local area around the seed. The

experimental result shows that weight decreasing with time has more potential. The updating rule is as follows.

$$V_{id}^{new} = W \cdot V_{id}^{old} + c_1 \cdot rand_1 \cdot (P_{id} - X_{id}) + c_2 \cdot rand_2 \cdot (P_{gd} - X_{id}) \quad (3)$$

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new} \quad (4)$$

where W is an inertia weight.

Shi and Ebberhart [48] subsequently explored what impact the inertia weight and maximum velocity have on the performance of the particle swarm optimizer, and provided guidelines for selecting these two parameters.

When we lack knowledge regarding the selection of V_{max} , it is also a good choice to set V_{max} equal to X_{max} . Furthermore, if a time-varying inertia weight is employed, even better performance can be expected. Because particle swarm optimization originated from efforts to model social systems, a thorough mathematical foundation of the methodology was not developed along with the algorithm. Recent work done by Clerc indicates that the use of a *constriction factor* may be necessary to ensure convergence of the PSOA [11]. The updating rule is as follows.

$$V_{id}^{new} = K \times [V_{id}^{old} + c_1 \times rand_1 \times (P_{id} - X_{id}^{old}) + c_2 \times rand_2 \times (P_{gd} - X_{id}^{old})] \quad (5)$$

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new} \quad (6)$$

$$K = \frac{2}{2 - \varphi - \sqrt{\varphi^2 - 4\varphi}}, \quad \varphi = c_1 + c_2 \text{ and } \varphi > 4 \quad (7)$$

where K is a constriction factor.

If particles' previous velocities are very close to zero, then all the particles will stop moving once they catch up with the global best particle, which may lead to premature convergence of the algorithm. Thus van den Bergh and Engelbrecht [4] suggest a new velocity update equation in order to keep the global best particle moving until it has reached a local minimum. The updating rule is as follows:

$$V_{id}^{new} = W \times V_{id}^{old} - X_{id}^{old} + P_{gd} + \rho^{old} \times (1 - 2rand_2) \quad (8)$$

$$X_{id}^{new} = W \times V_{id}^{new} + P_{gd} + \rho^{old} \times (1 - 2rand_2) \quad (9)$$

$$\rho_0 = 1.0, \quad \rho^{new} = \begin{cases} 2\rho^{old} & \text{if } \# \text{successes} > s_c \\ 0.5\rho^{old} & \text{if } \# \text{failures} > f_c \\ \rho^{old} & \text{otherwise} \end{cases} \quad (10)$$

where ρ is a scaling factor.

Social interaction is an important factor to improve PSOA performance. To enhance the social interactions in the algorithm, Zhao et al. [52] put forward a new method of improved PSOA. They propose the use of an improved adaptation strategy with enhanced social interactions for PSOA. This adaptation strategy uses the information of more particles to control the mutation operation. It also extends the original formulas of PSOA to search for the global optimal solution more effectively. This is similar to the human society in that a group of leaders can make better decisions. The updating rule is as follows:

$$V_{id}^{new} = W \times V_{id}^{old} + c_1 \times rand_1 \times (P_{id} - X_{id}^{old}) + \sum_{i=1}^n c_{2,i} \times rand_{2,i}(t) \times (\hat{P}_{i,t} - X_{id}^{old}) \quad (11)$$

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new} \quad (12)$$

$$W = w_{max} - \frac{w_{max} - w_{min}}{iter_{max}} \times iter \quad (13)$$

Xiao et al. [49] proposed a hybrid clustering approach based on a self-organizing map (SOM) neural network. The proposed algorithm uses a PSOA to evolve weights for the SOM neural network. The weights are trained by the SOM neural network in the first stage, and in the second stage they are optimized by the PSOA. The experimental results show that the hybrid method tries to tune the original SOM such that it can achieve a better tradeoff between the average quantization error and the topographic error. van der Merwe and Engelbrecht [44] proposed two new approaches using PSOA to cluster data. The first method, called the PSO clustering algorithm, shows how the PSOA can be used to find the centroids of a user-specified number of clusters. The second method, called hybrid PSOA, first uses K-means clustering to seed the initial swarm and then uses PSOA to refine the clusters formed by K-means. These new PSOAs were evaluated on six data sets, and compared with the performance of K-means clustering. Results show that both PSOAs have a great deal of potential. Chen and Ye also proposed a clustering analysis algorithm based on PSOA [10]. However, the concept behind their algorithm is similar to [44], and the main difference is in fitness function only. Finally, the experimental results obtained using four artificial data sets show the algorithm proposed in this study performs better than K-means and fuzzy C-means. Cohen and de Castro [12] presented a modified PSOA with self-organization in the updating rule for clustering analysis. In their PSOA, it is not necessary to calculate fitness value. The results show that it is better than the K-means method. Kuo et al. (2008) proposed a PSKO which combined PSO clustering [44] with K-means. The PSKO was evaluated on four data sets, and compared with the performance of K-means clustering, PSO clustering and hybrid PSO. The experimental results show that the PSKO algorithms outperform other algorithms.

2.3. Genetic algorithm (GA)

Holland [25] proposed Genetic Algorithm in "Adaptation in Natural and Artificial Systems" published in 1975, and its basic view is based on survival of the fittest in natural selection — Charles Darwin. Based on the survival and reproduction of the fittest, GA adopts a group of simulated encoded chromosomes and calculates the fitness of chromosomes. Each chromosome undergoes crossover and mutation to produce next generation. This evolution process continues until the stopping criteria are reached. GA is appropriate for large-sized, nonlinear space problems which solution is unpredictable. Relying on multi-point search and algorithmic features, it is not easy to fall into local optimal solution but can converge to universal optimal solution.

Since GA is good at searching, they are used to solve the clustering problem. Murthy and Chowdhury [43] proposed a GA-based method to solve the clustering problem and used three experiments on synthetic and real life data sets to evaluate the performance. The results show that the GA-based method may improve the final output of K-means. Al-Sultan and Maroof [2] studied several algorithms that include the K-means algorithm, the simulated annealing algorithm, the Tabu search algorithm, and the genetic algorithm and compare the performance for the clustering problems from the literature. In general, the result of the K-means algorithm is the worst. Krishna and Murty [31] proposed the GKA algorithm for clustering, and proved that the algorithm can converge to an optimal solution. In Ujjwal and Bandyopadhyay's experiments, the GA-clustering algorithm surpasses the K-means algorithm [42]. Also, Kuo et al. [33,36] presented different coding methods for genetic clustering. Above methods have a prerequisite that the clusters must be a fixed number, but in real world the number of clusters is unknown. Lin and Shiueng [40] propose a genetic-algorithm-based approach that can automatically find the proper number of clusters, and the recovery of their algorithm surpasses K-means with the IRIS data.

3. Methodology

This section presents the proposed order clustering system which consists of three components: (1) data collection and transformation,

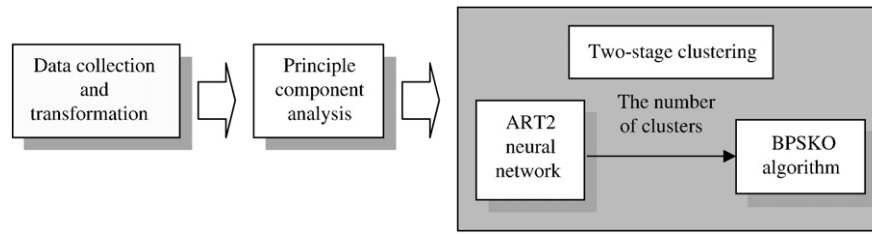


Fig. 1. Order clustering system.

(2) principle component analysis, and (3) clustering analysis, as shown in Fig. 1. The following subsections briefly discuss these three components.

3.1. Data collection and transformation

In order to classify the coming orders for their corresponding clusters, all previous orders must first be collected and clustered. These data can be retrieved from a company's enterprise resources planning system. Then, the bill of materials (BOM) for each order is used as the features for each order, or product.

3.2. Principle component analysis

Due to the vast number of materials used for each order, it is very time consuming to directly use these data for clustering analysis, and it is more feasible to reduce the dimensionality for these data in advance. A statistical method, principle component analysis, is employed for this purpose so that each order has a limited number of features. This can result in shorter computational time without influencing the computational outcome.

3.3. Clustering analysis

After implementing principle component analysis, the data obtained is applied for clustering analysis. Since the proposed clustering algorithm belongs to the supervised clustering method, an ART2 neural network is first employed to determine the number of clusters.

3.3.1. ART2 neural network

ART2 neural network architecture is designed for processing analog as well as binary input patterns [8]. An ART2 neural network consists of F1 and F2 layers. There are seven nodes in the F1 layer (W, X, U, V, P, Q). The input signal is processed by the F1 layer and then is passed from the bottom to the top value (b_{ij}). The result of the bottom-to-top value is an input signal of the F2 layer. The nodes of the F2 layer compete with each other to produce a winning unit and the winning unit returns the signal to the F1 layer. The match value is then

calculated with the top to bottom value (t_{ji}) in the F1 layer and compared with the vigilance value. If the match value is greater than the vigilance value, then the weight of b_{ij} and t_{ji} is updated. Otherwise, the reset signal is sent to the F2 layer and the winning unit is inhibited. After inhibition, the other winning unit is found in the F2 layer. If all of the F2 layer nodes are inhibited, the F2 layer produces a new node and generates the initial corresponding weights to the new node.

3.3.2. Hybrid of genetic algorithm and particle swarm optimization algorithm (HGAPSOA) for clustering

PSOA, like GA, is a population-based stochastic search process. The algorithm maintains a population of particles, where each particle represents a potential solution to an optimization problem. Thus, we propose a new technique, a hybrid of genetic algorithm and particle swarm optimization algorithm (HGAPSOA) for clustering problems. The HGAPSOA integrates particle swarm optimization, genetic algorithm and the K-means method.

The HGAPSOA randomly initializes a population of size $2N$. These individuals may be regarded as chromosomes in the case of GA, or as particles in the case of PSO. $2N$ individuals are fed into the real-coded GA to create $2N$ new individuals by reproduction, crossover and mutation operators. Then the $2N$ individuals are sorted by fitness, and the top N individuals are applied to *Improved PSO*. Thus *Improved PSO* can use better-performing individuals as the initial population to search for the optimum solution. In addition, the individuals with poor performance remain in order to escape the solution and avoid premature convergence. At the end of the HGAPSOA, the K-means algorithm is used for the purpose of fast convergence. The framework of the HGAPSOA is represented as shown in Fig. 2.

Improved PSO in Fig. 2 is a modified particle swarm optimization algorithm based on NPSO [39]. In *Improved PSO*, there is a crossover of P_{id} and P_{gd} for every particle. Then we can get two child particles. After comparison, let the particle with the smaller fitness value be the final child particle. Therefore, *Improved PSO* can use better-performing P_{id} and P_{gd} to lead all particles to search for the optimum solution. Then, the V_{id} is updated according to Eq. (14):

$$V_{id}^{new} = W * V_{id}^{old} + c_1 * rand_1 * (P_{id} - X_{id}) + c_2 * rand_2 * (P_{gd} - X_{id}), \quad (14)$$

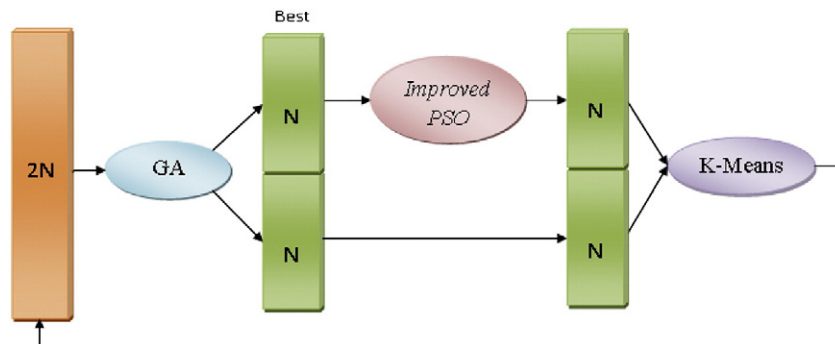


Fig. 2. The framework of the HGAPSOA.

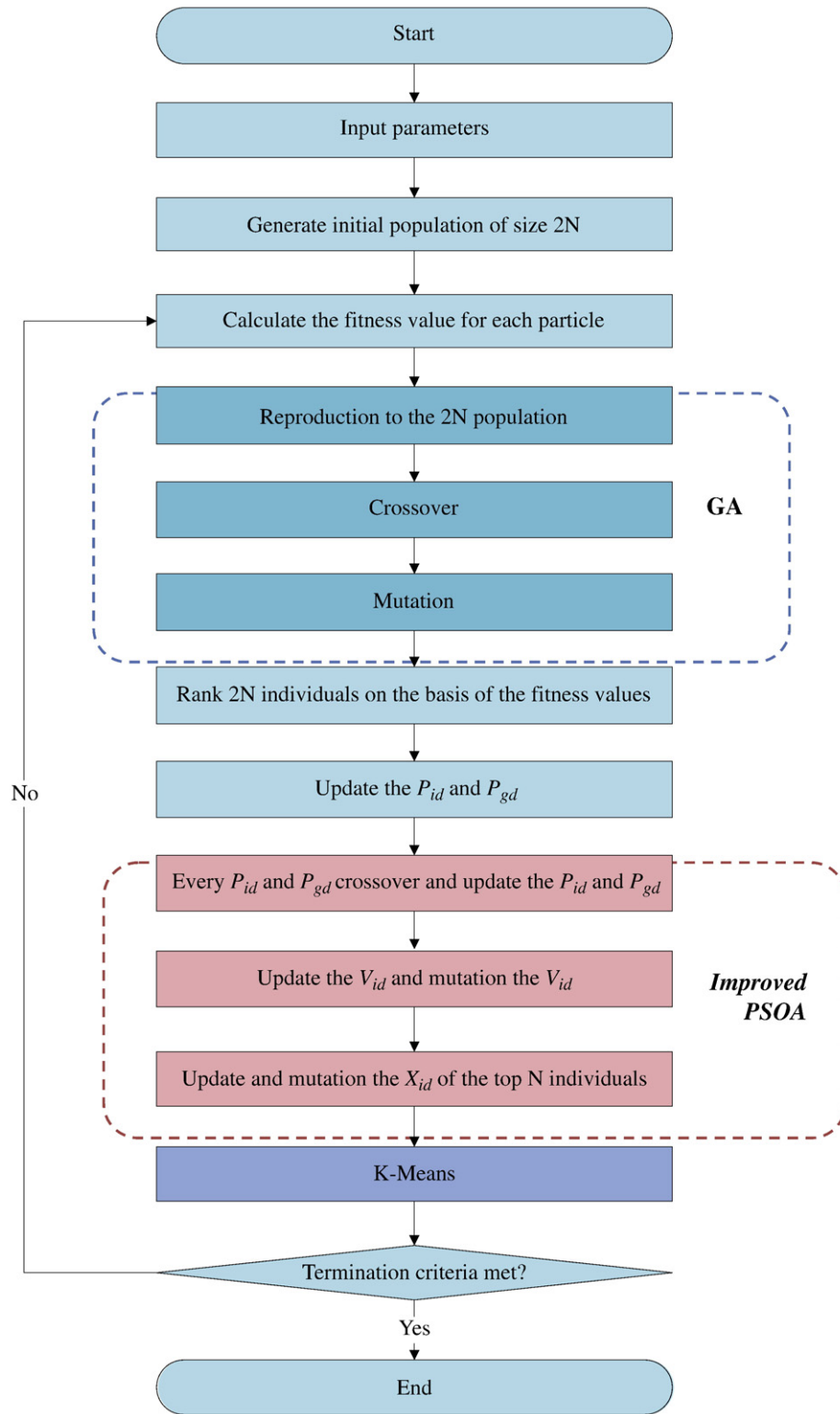


Fig. 3. HGAPSOA computational flow chart.

and mutated. Then X_{id} can be obtained using Eq. (15) as shown below:

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new}. \quad (15)$$

The next section will introduce the procedure to the clustering problem.

3.3.3. HGAPSOA computational procedures

The computational procedures for the proposed HGAPSOA are as follows:

Step 1 Set up parameters including population size $2N$ (number of particles), inertia weight, W , maximum velocity, V_{max} , and two learning factors, c_1 and c_2 .

Table 1

The corresponding information for each data set.

Name	Number of data	Number of features	Number of clusters
Iris	150	4	3
Wind	178	13	3
Vowel	990	10	11
Glass	214	9	6

Step 2 Initialize each particle randomly with initial position, X_{id} and velocity, V_{id} . In PSO clustering, we need to know the number of clusters, k , in advance. In this study, an ART2 neural network will provide this information to HGAPSOA. Since each particle is a vector containing k cluster centroids, each cluster's position, X_{id} can be represented as:

$$X_{id} = (x_{i1}, \dots, x_{ij}, \dots, x_{ik}) \quad (16)$$

where x_{ij} denotes the j th cluster's centroid for the i th particle.

Step 3 Calculate fitness value for each particle in order to adjust suitabilities.

$$\text{Fitness value} = \sum_{j=1}^k \left| \sum_{x \in c_{ij}} \|x - z_{ij}\| \right| \quad (17)$$

where

k denotes the cluster number

x denotes data vectors to be clustered

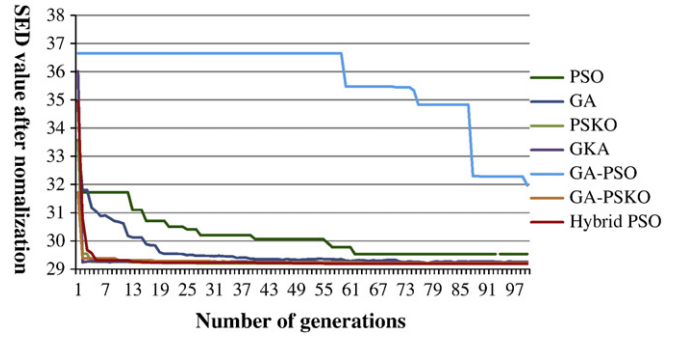
c_{ij} denotes the number of data vectors of the i th particle in cluster j and

$\|x - z_{ij}\|$ denotes the Euclidean distances between data vectors and cluster centroids

Table 2

The parameter setup for each algorithm.

PSOA related parameter setup	
	PSO clustering, PSKO algorithm, GA-PSO algorithm, and GA-PSKO algorithm
Number of particles	20
Number of generations	100
W	1.2 decrease to 0.4
c_1 and c_2	2
$X_{\max}$	Yes
$V_{\max}$	Yes
GA related parameter setup	
	GA algorithms, GKA algorithm, GA-PSO algorithm, and GA-PSKO algorithm
Number of particles	20
Number of generations	100
Crossover rule	100% (one-point crossover)
Mutation probability	5%
HGAPSOA	
Number of particles	10
Number of generations	100
W	1.2 decrease to 0.4
c_1 and c_2	2
$X_{\max}$	Yes
$V_{\max}$	Yes
Crossover rule	100% (one-point crossover)
Mutation probability	5%

**Fig. 4.** The converged curves for Iris data.

(3a) Calculate the Euclidean distance $d(x, z_{ij})$ to all cluster centroids according to the equation below:

$$d(x, z_{ij}) = \|x - z_{ij}\| \quad (18)$$

(3b) Assign x to cluster z_{ij} such that $d(x, z_{ij}) = \min_{c_1, \dots, c_k} \{d(x, z_{ij})\}$.

(3c) Calculate the fitness using Eq. (17)

Step 4 Apply real-coded GA operators (reproduction, crossover and mutation) to the $2N$ population and create another $2N$ population.

(4a) *Reproduction*: From the population, select the $2N$'s best individuals according to fitness.

(4b) *Crossover*: Apply two-parent crossover to update the $2N$ individuals.

(4c) *Mutation*: Apply mutation with 5% mutation probability to the $2N$ chromosomes according to the equation below:

$$x' = x + \text{rand} \times N(0, 1) \quad (19)$$

Step 5 Evaluate the fitness of each of the $2N$ individuals. Rank them on the basis of the fitness values.

Step 6 Update the global best (P_{gd}) and local best positions (P_{id}).

Step 7 *Crossover*: Based on NPSO [39], we apply every P_{id} and P_{gd} crossover and get two child particles, compare them and let the particle with smaller fitness value be the final child of the predecessors.

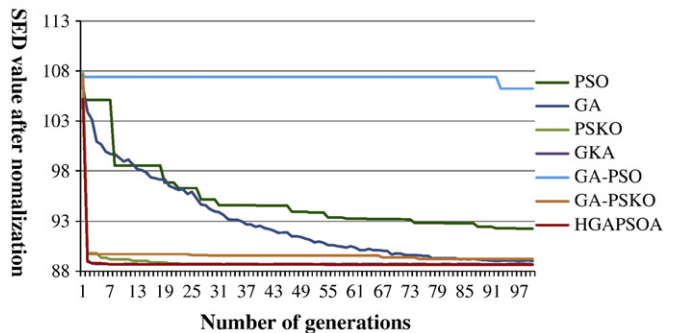
Step 8 (8a) Update the velocity (V_{id}) using Eq. (20).

$$V_{id}^{\text{new}} = W \times V_{id}^{\text{old}} + c_1 \times \text{rand}_1 \times (P_{id} - X_{id}^{\text{old}}) + c_2 \times \text{rand}_2 \times (P_{gd} - X_{id}^{\text{old}}) \quad (20)$$

where

c_1 and c_2 are two positive constant

rand_1 and rand_2 are two random functions in the range [0,1] and

**Fig. 5.** The converged curves for Wine data.

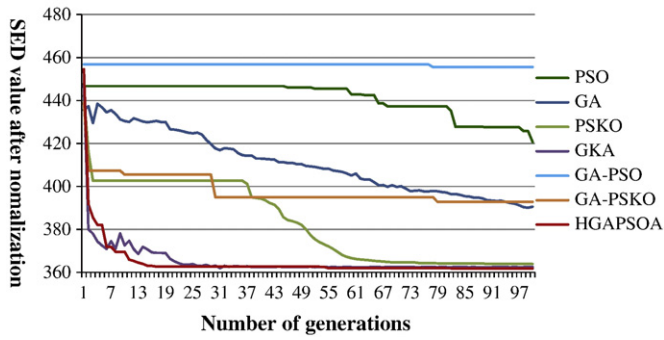


Fig. 6. The converged curves for Vowel data.

W is the inertia weight.

(8b) *Mutation*: Update the V_{id} utilizing Eq. (21).

$$V_{old}^{new'} = V_{old}^{new} + rand \times N(0, 1) \quad (21)$$

Step 9 (9a) Update the position vector (X_{id}) of the top N individuals using Eq. (22).

$$X_{id}^{new} = X_{id}^{old} + V_{id}^{new} \quad (22)$$

(9b) *Mutation*: Update the X_{id} of the top N individuals utilizing Eq. (23).

$$X_{id}^{new'} = X_{id}^{old} + rand \times N(0, 1) \quad (23)$$

Step 10 Apply K-means operators to the $2N$ population and create another $2N$ population using Eq. (24).

$$z_{ij} = \frac{1}{c_{ij}} \sum_{x \in c_{ij}} x \quad (24)$$

Step 11 If one of the stopping criteria is satisfied then go to Step 12. Otherwise, go to Step 3.

Step 12 Output the particle with the minimum fitness value in the last generation.

Fig. 3 illustrates the computation flowchart of HGAPSOA.

3.3.4. Algorithm validation

To examine HGAPSOA's performance, this study compares various clustering methods with the proposed clustering algorithm. These algorithms include GA [42], GKA [31], PSO [44], PSKO (Kuo et al.,

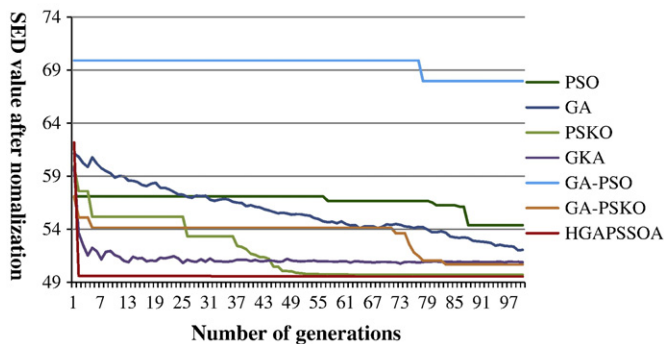


Fig. 7. The converged curves for Glass data.

Table 3
Clustering results for seven algorithms.

Average results of SED for 30 runs														
Data set	Algorithm													
GA	GKA		PSO		PSKO		GA-PSO		GA-PSKO		HGAPSOA			
	Average	Best	Average	Best	Average	Best	Average	Best	Average	Best	Average	Best		
Iris	29.835 + 1.657	29.180	29.262 + 0.016	29.235	29.790 + 0.240	29.523	29.206 + 0.006	29.128	35.375 + 2.032	31.682	29.226 + 0.011	29.212	29.190 + 0.008	29.177
Glass	52.465 + 1.288	50.529	51.769 + 2.323	49.859	57.485 + 2.045	54.089	50.021 + 0.738	49.604	59.132 + 1.718	56.633	51.268 + 0.816	50.378	49.710 + 0.286	49.479
Vowel	391.611 + 5.155	380.665	363.487 + 1.240	361.786	421.443 + 6.035	410.691	386.578 + 21.567	362.431	446.479 + 6.498	428.878	394.069 + 7.728	377.568	362.449 + 0.788	361.652
Wine	93.129 + 3.734	89.862	88.712 + 0.007	88.695	97.885 + 5.242	92.236	88.695 + 4.34E-14	88.695	111.943 + 4.948	104.388	88.695 + 4.336E-14	88.695	88.671 + 0.009	88.649

Table 4

The multivalence test in Iris data set.

Data set: Iris	HGAPSOA vs. PSO	HGAPSOA vs. PSKO	HGAPSOA vs. GA	HGAPSOA vs. GKA	HGAPSOA vs. GA-PSO	HGAPSOA vs. GA-PSKO
Mann–Whitney U	.000	51.000	64.000	.000	.000	.000
Wilcoxon W	465.000	516.000	529.000	465.000	465.000	465.000
Z	−6.653	−5.913	−5.707	−6.661	−6.653	−6.656
Asymp.sig.(2-tailed)	.000	.000	.000	.000	.000	.000

2008), GA-PSO [13] and GA-PSKO. GA-PSKO is a clustering algorithm which combines GA-PSO with K-means.

4. Simulation

This section applies some benchmark data sets to assess the proposed algorithm, HGAPSOA, in comparison with GA, GKA, PSO, PSKO, GA-PSO and GA-PSKO. A detailed discussion is presented as follows.

4.1. Data set

This section adopts four data sets provided by the website of the Department of Information and Computer Science, University of California (<http://archive.ics.uci.edu/ml/datasets.html>). Table 1 lists the corresponding information including data set name, number of data, number of features, and number of clusters, for each data set.

4.2. Data preprocessing and parameter setup

The features in each data set may have different value ranges. If these ranges are quite different from each other, then even a good clustering algorithm still cannot obtain good clustering results. Thus, this study normalizes raw data into the same value range. The basic goal of normalization is to find the largest and smallest values for each feature. And then each data in the feature is divided by the difference of the largest and smallest values. Basically, this is a very common normalization method.

It is necessary to determine the parameter values for our proposed HGAPSOA and other clustering algorithms. We can easily classify them into three categories, PSO-based algorithm, GA-based algorithm and HGAPSOA. The corresponding information is summarized in Table 2.

4.3. Algorithm evaluation indicator

Although mean square error (MSE), or within variance, is a commonly used indicator for evaluating clustering efficiency, we employed the sum of Euclidean distances (SED), as presented by Kaufman and Rousseuw [29], since they showed that SED is more robust than MSE in evaluating clustering results. Basically, the smaller the SED is, the better the result, and SED is represented as:

$$SED = \sum_{i=1}^n \sum_{j=1, x_i \in C_j}^k d(x_i, m_j) \quad (25)$$

where

n total number of data,
 C_j the j th cluster,

K the number of clusters,
 D Euclidean distance,
 x_i the dimension of the i th data, and
 m_j the coordinate of the j th cluster's center.

4.4. Result analysis

4.4.1. Converge speed analysis

Figs. 4 to 7 illustrate the converged curves for four data sets, respectively. The proposed method integrates the *Improved PSO*, GA and K-means algorithm. Thus, *Improved PSO* has better-performing P_{id} and P_{gd} which leads all particles to search for the optimum solution. Moreover, the K-means converges fast. As a result, HGAPSOA can converge without exceeding 20 iterations. Also, it can converge faster than the other six methods, GA, GKA, PSO, PSKO, GA-PSO and GA-PSKO.

4.4.2. Clustering accuracy analysis

In this study, we select SED as the performance indicator and a smaller SED value indicates better performance. In addition, each experiment is conducted thirty times and the average and standard deviation are calculated as the final result. The average SED and the corresponding standard deviation for each experiment are listed in Table 3.

Among these seven clustering algorithms, the proposed method, HGAPSOA, has the best performance in four data sets, including iris, wine, vowel, and glass. By examining the computational results for thirty runs, we found that HGAPSOA overcomes the problem of premature convergence.

4.4.3. Multivalence test

In order to verify the difference of the average value between HGAPSOA and other clustering algorithms, we apply Mann–Whitney U to test the difference at a 95% confidence level. The null hypothesis is listed as follow:

Ho The average value of two independent samples has no difference.

H1 The average value of two independent samples has difference.

The Cochran test statistic is as follows:

$$Z = \frac{U - \frac{n_1(n_1 + n_2 + 1)}{2}}{\sqrt{\frac{1}{12}n_1n_2(n_1 + n_2 + 1)}}, \quad U = |U_1, U_2| \quad (26)$$

where $U_1 = n_1n_2 - \frac{n_1(n_1 + 1)}{2} - W_1$ and $U_2 = n_1n_2 - \frac{n_2(n_2 + 1)}{2} - W_2$.

The test results are listed in Tables 4 to 7 and show that HGAPSOA mostly shows the difference ($P < 0.05$). Thus we could conclude that HGAPSOA has more remarkable performance than other methods.

Table 5

The multivalence test in Glass data set.

Data set: Glass	HGAPSOA vs. PSO	HGAPSOA vs. PSKO	HGAPSOA vs. GA	HGAPSOA vs. GKA	HGAPSOA vs. GA-PSO	HGAPSOA vs. GA-PSKO
Mann–Whitney U	.000	234.000	2.000	44.000	.000	3.000
Wilcoxon W	465.000	699.000	467.000	509.000	465.000	468.000
Z	−6.653	−3.194	−6.623	−6.003	−6.653	−6.609
Asymp.sig.(2-tailed)	.000	.000	.000	.000	.000	.000

Table 6

The multivalence test in vowel data set.

Data set: Vowel	HGAPSOA vs. PSO	HGAPSOA vs. PSKO	HGAPSOA vs. GA	HGAPSOA vs. GKA	HGAPSOA vs. GA-PSO	HGAPSOA vs. GA-PSKO
Mann–Whitney <i>U</i>	.000	20.000	.000	196.000	.000	.000
Wilcoxon <i>W</i>	465.000	485.000	465.000	661.000	465.000	465.000
<i>Z</i>	−6.653	−6.357	−6.653	−3.755	−6.653	−6.653
Asymp.sig.(2-tailed)	.000	.000	.000	.000	.000	.000

5. Model evaluation results

Section 4 has demonstrated the capability of the proposed HGAPSOA to outperform GA, GKA, PSO, PSKO, GA-PSO and GA-PSKO. This section applies HGAPSOA for order clustering in order to reduce SMT production system setup time. Advantech Company provided the related data for assessment. The setup of production line-change for different orders has become a critical bottleneck in SMT production systems because it is very time consuming in production line-change to implement material-preparation and material-binding tasks by hand. Failure to complete this process prior to the start of production of the next order causes machine idle time and results in low production capability utilization. Therefore, we apply a two-stage clustering method [34] to group similar products or orders together and find the same materials (shared materials) that will be used for all the products in the same group. This enables the production engineers to put the same needed materials in the same material positions for SMT retrieval equipment, so production engineers can arrange for similar products to be produced together. Thus, the material-preparation operation can be simplified in order to reduce SMT setup time.

5.1. Order clustering system framework

The main procedures of the order clustering system framework are illustrated in Fig. 8.

A material report is provided by an industrial PC company. Using ACCESS SQL query, there are a total of 2826 different products, or orders, and a total of 2516 materials used. In order to cluster these products, it is necessary to filter the noise of the products and material data and then transform them into a 2826×2516 two-dimensional matrix.

5.2. Data preprocessing

Since there are 2516 features for the product feature matrix, it is very time consuming for clustering analysis with so many features. In order to reduce the computational time while also maintaining solution quality, this study uses Matlab 7.0 to make the principle component analysis for the product feature matrix in order to extract the dimensions for the principle component factors. The selection of factors is based on the Eigenvalue which should be larger than 1. After analysis, there are a total of 200 principle component factors. The cumulative explanation variance is 81.98%.

Principle component analysis reduces the original feature matrix to a size of 2826×2516 to 2826×200 . The proposed method, ART2+ HGAPSOA algorithm can be applied for clustering analysis. These two parts are shown in the following subsections.

Table 7

The multivalence test in Wine data set.

Data set: Wine	HGAPSOA vs. PSO	HGAPSOA vs. PSKO	HGAPSOA vs. GA	HGAPSOA vs. GKA	HGAPSOA vs. GA-PSO	HGAPSOA vs. GA-PSKO
Mann–Whitney <i>U</i>	.000	.000	.000	.000	.000	.000
Wilcoxon <i>W</i>	465.000	465.000	465.000	465.000	465.000	465.000
<i>Z</i>	−6.654	−7.112	−6.653	−6.655	−6.653	−7.112
Asymp.sig.(2-tailed)	.000	.000	.000	.000	.000	.000

5.3. Clustering analysis

In order to apply our proposed clustering algorithm, HGAPSOA, it is necessary to know the number of clusters in advance. Based on our previous research [34], this study applies a two-stage clustering method, ART2+ HGAPSOA which is a kind of method that processes gray value in the first stage, ART2 automatically finds the number of clusters, while the second stage uses the HGAPSOA algorithm to find the final solution.

5.3.1. Determination of cluster number

The ART2 algorithm was coded by using Matlab 7.0. Since the number of clusters in ART2 algorithm is determined by the vigilance value, once it is well determined, ART2 can automatically cluster the data. Thereafter, the second stage employs GA, GKA, PSO, PSKO, GA-PSO, GA-PSKO and HGAPSOA to find the final solutions. Since the vigilance value dramatically affects the clustering outcome, this study uses Wilk's Lambda value as the indicator for determining the number of clusters. Wilk's Lambda is frequently applied by MANOVA as the indicator for determining the number of clusters. If Wilk's Lambda value suddenly increases in two different numbers of clusters, then the number of clusters prior to variance can be treated as the best number of clusters. Theoretically, Wilk's Lambda value is defined as:

$$\text{Wilk's Lambda} = \frac{SS_{\text{within}}}{SS_{\text{total}}} \quad (27)$$

where SS_{within} and SS_{total} are the within-cluster and total variances, respectively.

We set up different vigilance values to obtain different cluster numbers from 30 to 80. Fig. 9 depicts the corresponding Wilk's Lambda values for different vigilance values. If the incremental range is 10, cluster numbers from 50 to 60 have sudden variance in Wilk's Lambda value. For further investigation, from cluster 55 to 57, there is also similar phenomenon. Thus, cluster number 57 is the best number of clusters.

5.3.2. Comparison of different clustering algorithms

Based on the ART2 algorithm's result, GA, GKA, PSO, PSKO, GA-PSO, GA-PSKO and HGAPSOA are further used to find the final solution for comparison. Table 8 depicts the sum of Euclidean distances (SEDs) of these algorithms. Each algorithm is run ten times and average and standard deviation are calculated, respectively. Table 8 indicates that the ART2+ HGAPSOA has the smallest SED value, 2234.629.

5.4. Shared materials

According to the clustering results of ART2+ HGAPSOA, the products are grouped into fifty seven clusters. The corresponding

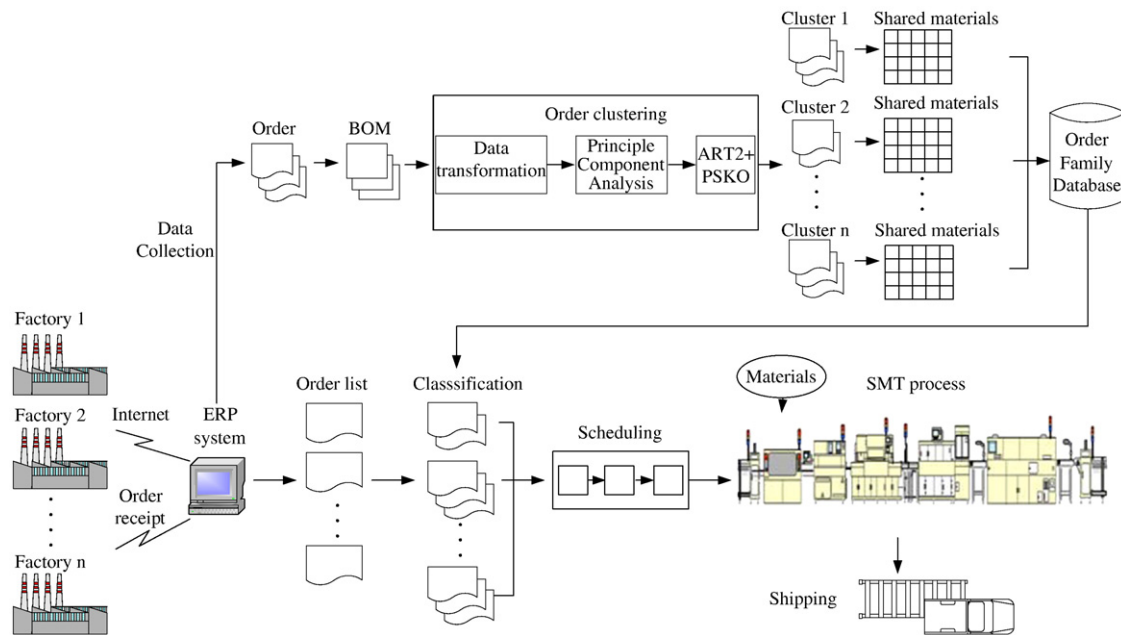


Fig. 8. Order clustering system.

number of products and number of shared materials for these clusters with the number of shared materials over one hundred are presented in Table 9.

5.5. Order classification

A feedforward neural network with an error backpropagation learning algorithm will learn the clustering results from ART2+ HGAPSOA. Since there are a total of 2516 data, they are divided into 1696 training samples, 564 testing samples, and 565 evaluation samples. The network topology includes 200 input nodes and 57 output nodes, respectively. Regarding the number of hidden nodes, trial and error method is applied. After testing, the hidden layer with 179 nodes has the lowest mean square error as the training rate and momentum rate are 0.5 and 0.9, respectively.

5.6. Performance evaluation for production line-changes

This study simulates production line-change efficiency through the on-field collection of the production plan. In order to simplify the variables which may occur in the practical applications, this study makes the following assumptions:

1. There is no stock-out condition.
2. The machines do not have problems of shutdown or malfunction.

3. There is no human error (e.g. material-binding error or material-preparation miss) which may affect the production plan during the material-preparation process of material-preparation work.

Through job-shop measuring, the material-binding per material is 50 s, while material-preparation time is 0.5 h. Table 10 lists the production plan for the SMT production line from May 4, 2005 to May 5, 2005. These orders' corresponding clusters are listed as well.

According to our assumptions, the actual production time consists of two parts: material-binding time and mounting time. In addition, it should be noted that all the boards produced by the SMT need to first implement the material-binding operation, which is then followed by the mounting operation. In other words, if the material-binding operation is completed as the next product is going to be produced, then the product can be smoothly mounted; otherwise, mounting cannot be started until the material-binding operation is completed. However, this would result in machine idle time.

Next, two scheduling rules, first-come first-served (FCFS) and shortest processing time (SPT), are applied for production scheduling in order to evaluate the proposed ART2+ HGAPSOA's performance.

5.6.1. Production model before improvement

5.6.1.1. FCFS. Using the FCFS rule for scheduling the production process before improvement, the material-preparation time is 23.9 h, while the total production time is 34.3 h. In addition, the total machine idle time is $2.8(0.5 + 0.7 + 1.6)$ h.

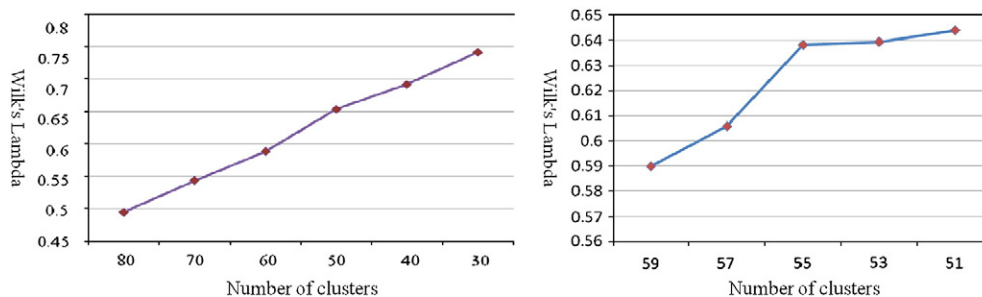


Fig. 9. Wilk's Lambda values.

Table 8

The SED values for every clustering algorithms.

Average and standard deviation results of SED For 10 runs							
Algorithms	ART2 ± GA	ART2 ± GKA	ART2 ± PSO	ART2 ± PSKO	ART2 ± GA-PSO	ART2 ± GA-PSKO	ART2 ± HGAPSOA
Average and standard deviation	3403.891 ± 103.856	2247.473 ± 30.480	2336.318 ± 16.589	2331.605 ± 19.064	2321.627 ± 19.511	2329.585 ± 17.657	2234.629 ± 18.606
Best solution	3305.202	2209.569	2308.792	2295.650	2284.499	2300.389	2201.899

5.6.1.2. SPT. Similarly, using the SPT rule to schedule the production process before improvement, the material-preparation time is 23.9 h. Total production time and total machine idle time are 34.9 h and 5.6 (0.5 + 0.6 + 2.1 + 1.1 + 0.8 + 0.5) h, respectively.

5.6.2. Production model after improvement

This subsection employs ART2+ HGAPSOA to classify the orders first and arrange that the orders belonging to the same cluster are scheduled together. Table 9 indicates that cluster 2 and cluster 24 have 111 shared materials and 183 shared materials. Due to effect of shared materials, the shared materials only need to be prepared once. This can save material-preparation time and reduce the SMT line-change time.

5.6.2.1. FCF. Using the FCFS rule to schedule the production process after improvement, the material-preparation time is 19.7 h, while the total production time and total machine idle time are 31.5 h and 0 h.

5.6.2.2. SPT. The material-preparation time, total production time and total machine idle time are 19.9 h, 31.3 h, and 2(0.1 + 1 + 0.3 + 0.6)h, respectively.

For the purpose of comparison, the above results are summarized in Table 11. This indicates that the material-preparation time is reduced from 23.9 h to 19.9 h. This is a 16.74% decrease. In regard to total production time, there are 8.16% and 10.32% decreases with the FCFS rule and the SPT rule, respectively. The effect on total machine idle time is especially dramatic. The FCFS rule improved from 2.8 h to 0 h, while it improved from 5.6 h to 2 h for SPT. Thus, no matter what kind of scheduling rules is employed, using clustering analysis to arrange similar orders together really does save production time and idle time.

6. Conclusions

This study has proposed a two-stage clustering algorithm, ART2+ HGAPSOA, for clustering analysis. The simulation results have shown that HGAPSOA outperforms GA, GKA, PSO, PSKO, GA-PSO and GA-PSKO algorithms in four data sets.

Table 9

Some clusters and their corresponding numbers of shared materials.

Cluster	Number of products	Number of shared materials	Cluster	Number of products	Number of shared materials
2	16	111	30	5	133
3	6	112	31	18	158
6	6	101	32	2	157
8	8	110	34	13	148
15	3	155	37	13	128
16	8	158	38	38	106
18	11	134	39	36	101
21	30	143	44	12	106
24	6	183	45	6	117
25	15	112	48	16	120
26	35	130	51	38	131
27	32	109	57	21	137

Table 10

Two-day production plan of SMT production system.

Original production system				Improved production system			
A	B	C	D	E	F	G	H
1	9498968300	70	4.9 = 4.4 + 0.5	4.4	27	188	2.6 = 188 × (50/60/60)
2	9496064000	10	1.2 = 0.7 + 0.5	0.7	27	204	2.8 = 204 × (50/60/60)
3	9496074410	30	2.6 = 2.1 + 0.5	2.1	38	205	2.8 = 205 × (50/60/60)
4	9496074430	15	1.5 = 1 + 0.5	1	38	198	2.8 = 198 × (50/60/60)
5	9498968000	50	3.6 = 3.1 + 0.5	3.1	27	186	2.6 = 186 × (50/60/60)
6	9493650000	96	1.3 = 0.8 + 0.5	0.8	17	26	0.4 = 26 × (50/60/60)
7	9493650010	96	2.5 = 2 + 0.5	2	17	63	0.9 = 63 × (50/60/60)
8	9498968600	35	1.6 = 1.1 + 0.5	1.1	1	95	1.3 = 95 × (50/60/60)
9	9496687102	40	2.5 = 2 + 0.5	2	28	149	2.1 = 194 × (50/60/60)
10	9496074210	110	8.1 = 7.6 + 0.5	7.6	14	205	2.8 = 205 × (50/60/60)
11	9496074230	60	4.6 = 4.1 + 0.5	4.1	14	201	2.8 = 201 × (50/60/60)

A: Production sequence.

B: Product type.

C: Quantity.

D: Predicted labor time (h) = mounting time + material-preparation time.

E: Mounting time (h).

F: Cluster belonged.

G: Number of materials used.

H: Material-binding time (h) = number of materials used × per material-binding time.

Table 11

Model evaluation results.

	Scheduling rule	Material-preparation time (h)	Total production time (h)	Machine idle time (h)
FCFS	Before improvement	23.9	34.3	2.8
	After improvement	19.9	31.5	0
SPT	Before improvement	23.9	34.9	5.6
	After improvement	19.9	31.3	2

The proposed method was also applied to cluster orders to improve production performance. The model evaluation results show that both production time and machine idle time can be reduced since similar products are scheduled for production together. The production engineers can simplify the binding process since only different materials in the same cluster should be bound during product change. The results show that the machine idle time decreases from 2.8 h to 0 h for the FCFS rule and from 5.6 h to 2 h for SPT rule. In addition, total production time has around 8.16% decrease for FCFS rule and 10.32% decrease for the SPT rule. Thus, if the proposed method can be integrated with other advanced scheduling methods, the improvements in performance can be more significant.

A possible direction for future research would be to adopt different updating rules in order to get more adequate results. Furthermore, combining Tabu search clustering or simulated annealing clustering algorithms may be feasible. From the two-stage point of view, other clustering algorithms like the adaptive double SOM network can be used to decide the number of clusters. The proposed algorithm can also be utilized for many different areas of applications, like customer relationship management, market segmentation, and others.

Acknowledgements

This study was supported by the National Science Council of the Taiwan Government under contract number NSC 96-2416-H-011-018-MY3. The support is greatly appreciated. In addition, acknowledgements are owed to the case company for providing the related data.

References

- [1] K. Al-Sultan, A Tabu search approach to the clustering problem, *Pattern Recognition* 28 (9) (1995) 1443–1451.
- [2] Khaled S. Al-Sultan, Khan M. Maroof, Computational experience on four algorithms for the hard clustering problem, *Pattern Recognition Letters* 17 (3) (1996) 295–308 March 6.
- [3] G. Ball, D. Hall, A clustering technique for summarizing multivariate data, *Behavioral Science* 12 (1967) 153–155.
- [4] F. van den Bergh, A.P. Engelbrecht, A new locally convergent particle swarm optimiser, *Proceedings of IEEE International Conference on Systems, Man and Cybernetics*, 3, 2002.
- [5] J. Bezdek, A convergence theorem for the fuzzy ISO-DATA clustering algorithm, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 2 (1980) 1–8.
- [6] J. Bezdek, R. Hathaway, Numerical convergence and interpretation of the fuzzy c-shells clustering algorithms, *IEEE Transactions on Neural Networks* 3 (5) (1992) 787–793.
- [7] G. Carpenter, S. Grossberg, A massively parallel architecture for a self-organizing neural pattern recognition machine, *Computer Vision, Graphics, and Image Processing* 37 (1987) 54–115.
- [8] G. Carpenter, S. Grossberg, ART2: self-organization of stable category recognition codes for analog input patterns, *Applied Optics* 26 (23) (1987) 4919–4930.
- [9] G. Carpenter, S. Grossberg, D. Rosen, Fuzzy ART: fast stable learning and categorization of analog patterns by an adaptive resonance system, *Neural Networks* 4 (1991) 759–771.
- [10] C.Y. Chen, F. Ye, Particle swarm optimization algorithm and its application to clustering analysis, *IEEE International Conference on Networking, sensing and Control* (2004) 789–794.
- [11] M. Clerc, The swarm and the queen: towards a deterministic and adaptive particle swarm optimization, *Evolutionary Computation* (1999) 1951–1957.
- [12] S.C.M. Cohen, L.N. de Castro, Data clustering with particle swarms, *IEEE Congress on Evolutionary Computations* (2006) 1792–1798.
- [13] S. Du, W. Li, K. Cao, A learning algorithm of artificial neural network based on GA-PSO, *Proceedings of the 6th World Congress on Intelligent Control and Automation*, 2006 Dalian, China.
- [14] R. Eberhart, J. Kennedy, A new optimizer using particle swarm theory, *Proceedings of the Sixth International Symposium on Micro Machine and Human Science*, 1995, pp. 39–43.
- [15] V. Estivill-Castro, I. Lee, AMOEBA: hierarchical clustering based on spatial proximity using Delaunay diagram, *Proceedings 9th International Spatial Data Handling (SDH2000)*, 2000, pp. 10–12.
- [16] V. Estivill-Castro, I. Lee, AUTOCLUST: automatic clustering via boundary extraction for massive point data sets, *Proceedings 5th International Conference Geo-Computation*, 2000, pp. 23–25.
- [17] E. Forgy, Clustering analysis of multivariate data: efficiency versus interpretability of classification, *Biometrics* 21 (1965) 768–769.
- [18] I. Gath, A. Geva, Unsupervised optimal fuzzy clustering, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11 (7) (1989) 773–781.
- [19] A.B. Geva, Hierarchical unsupervised fuzzy clustering, *IEEE Transactions on Fuzzy Systems* 7 (6) (1999) 723–733.
- [20] S. Guha, R. Rastogi, K. Shim, CURE: an efficient clustering algorithm for large databases, *Proceedings ACM SIGMOD International Conference Management of Data* (1998) 73–84.
- [21] S. Guha, R. Rastogi, K. Shim, ROCK: a robust clustering algorithm for categorical attributes, *Information Systems* 25 (5) (2000) 345–366.
- [22] L. Hall, I. Özyurt, J. Bezdek, Clustering with a genetically optimized approach, *IEEE Transactions on Evolutionary Computation* 3 (2) (1999) 103–112.
- [23] G. Hamerly, C. Elkan, Learning the K in K-means, *Proceedings of 7th Annual Conference on Neural Information Processing Systems*, 2003.
- [24] J. Han, M. Kamber, *Data Mining: Concepts and Techniques*, 2nd ed. Morgan Kaufmann, 2006.
- [25] J. Holland, *Adaptation in Natural and Artificial System*, University of Michigan Press, Ann Arbor, MI, 1975.
- [26] K. Huang, A synergistic automatic clustering technique (Syneract) for multispectral image analysis, *Photogrammetric Engineering Remote Sensing* 1 (1) (2002) 33–40.
- [27] A.K. Jain, M.N. Murty, P.J. Flynn, Data clustering: a review, *ACM Computing Surveys* 31 (3) (1999) 264–323.
- [28] G. Karypis, E. Han, V. Kumar, Chameleon: hierarchical clustering using dynamic modeling, *IEEE Computer* 32 (8) (1999) 68–75.
- [29] L. Kaufman, P. Rousseeuw, *Finding groups in data: an introduction to cluster analysis*, Wiley (1990).
- [30] T. Kohonen, The self-organizing map, *Proceedings IEEE* 78 (9) (1990) 1464–1480.
- [31] K. Krishna, M.N. Murty, Genetic K-means algorithm, *IEEE Transactions on Systems, Man, and Cybernetics* 29 (3) (1999) 433–439.
- [32] R. Krishnapuram, J. Keller, A possibilistic approach to clustering, *IEEE Transactions Fuzzy Systems* 1 (2) (1993) 98–110.
- [33] R.J. Kuo, K. Chang, S.Y. Chien, Integration of self-organizing feature maps and genetic-algorithm-based clustering method for market segmentation, *Journal of Organizational Computing and Electronic Commerce* 14 (1) (2004) 43–60.
- [34] R.J. Kuo, J.L. Liao, C. Tu, Integration of ART2 neural network and genetic K-means algorithm for analyzing Web browsing paths in electronic commerce, *Decision Support Systems* 40 (2) (2005) 355–374.
- [35] R.J. Kuo, H.S. Wang, T.L. Hu, S.H. Chou, Application of ant K-means on clustering analysis in data mining, *International Journal of Computers and Mathematics with Applications* 50 (2005) 1709–1724 November–December.
- [36] R.J. Kuo, Y.L. An, H.S. Wang, W.J. Chung, Integration of self-organizing feature maps neural network and genetic K-means algorithm for market segmentation, *Expert Systems with Applications* 30 (2) (February, 2006) 313–324.
- [37] Kuo, R.J., Wang, M.J., and Huang, T.W., “An application of particle swarm optimization algorithm to clustering analysis”, *Journal of Soft Computing*, 2010. (in press), doi:10.1007/S00500-009-00539-5.
- [38] C.Y. Lee, E.K. Antonsson, Dynamic partition clustering using evolution strategies, *The Third Asia-Pacific Conference on Simulated Evolution and Learning*, 2000.
- [39] Z. Lian, X. Gu, B. Jiao, A novel particle swarm optimization algorithm for permutation flow-shop scheduling to minimize makespan, *Chaos, Solitons & Fractals* 35 (5) (2008) 851–861.
- [40] Y.T. Lin, B.Y. Shiueng, A genetic approach to the automatic clustering problem, *Pattern Recognition* 34 (2) (2001) 415–424.
- [41] A. Lorente, X. Descombes, J. Zerubia, Fully unsupervised fuzzy clustering with entropy criterion, *International Conference on Pattern Recognition*, 3, 2000, pp. 3998–4001.
- [42] U. Maulik, S. Bandyopadhyay, Genetic algorithm-based clustering technique, *Pattern Recognition* 33 (9) (2000) 1455–1465.
- [43] C.A. Murthy, N. Chowdhury, In search of optimal clusters using genetic algorithms, *Pattern Recognition Letters* 17 (8) (1996) 825–832.
- [44] D.W. van der Merwe, A.P. Engelbrecht, Data clustering using particle swarm optimization, *The Congress on Evolutionary Computation* (2003) 215–220.
- [45] M.G.H. Omran, A. Salman, A.P. Engelbrecht, Dynamic clustering using particle swarm optimization with application in image segmentation, *Pattern Analysis and Applications* 8 (2006) 332–344.
- [46] D. Pelleg, A. Moore, X-means: extending K-means with efficient estimation of the number of clusters, *Proceedings of the 17th International Conference on Machine Learning*, Morgan Kaufmann, San Francisco, CA, 2000 pp. 727–734.
- [47] Y. Shi, R. Eberhart, A modified particle swarm optimizer, *Proceedings of the IEEE International Conference on Evolutionary Computation* (1998) 69–73.
- [48] Y. Shi, R. Eberhart, *Parameter Selection in Particle Swarm Optimization*, *Evolutionary Programming*, New York, 1998 pp. 591–600.
- [49] X. Xiao, E.R. Dow, R. Eberhart, Z.B. Miled, R.J. Oppelt, Gene clustering using self-organizing maps and particle swarm optimization, *Proceedings of the International Parallel and Distributed Processing Symposium*, 2003, pp. 22–28.
- [50] R. Xu, D. Wunsch, Survey of clustering algorithms, *IEEE Transactions on Neural Networks* 16 (3) (2005) 645–678.
- [51] T. Zhang, R. Ramakrishnan, M. Livny, BIRCH: an efficient data clustering method for very large databases, *Proceedings ACM SIGMOD Conference Management of Data* (1996) 103–114.
- [52] B. Zhao, C.X. Guo, B.R. Bai, Y.J. Cao, An improved particle swarm optimization algorithm for unit commitment, *International Journal of Electrical Power & Energy Systems* 28 (7) (2006) 482–490.



R.J. Kuo received the MS degree in Industrial and Manufacturing Systems Engineering from Iowa State University, Ames, IA, in 1990 and the PhD degree in Industrial and Management Systems Engineering from the Pennsylvania State University, University Park, PA, in 1994. Currently, he is a Professor in the Department of Industrial Management at National Taiwan University of Science and Technology, Taiwan. His research interests include architecture issues of computational intelligence and their applications in forecasting, electronic business, logistics, supply chain management and decision support systems.



L.M. Lin received the MS degree in Industrial Engineering and Management from National Taipei University of Technology, Taiwan. Currently, she is an engineer of AU Optronics Corporation, Taiwan. Her research interests include architecture issues of computational intelligence and their applications in production management.